

Measurement of Power and Energy

Exp:2.1: Measurement of Power

AIM

Three phase power measurement by two wattmeter method.

Theory

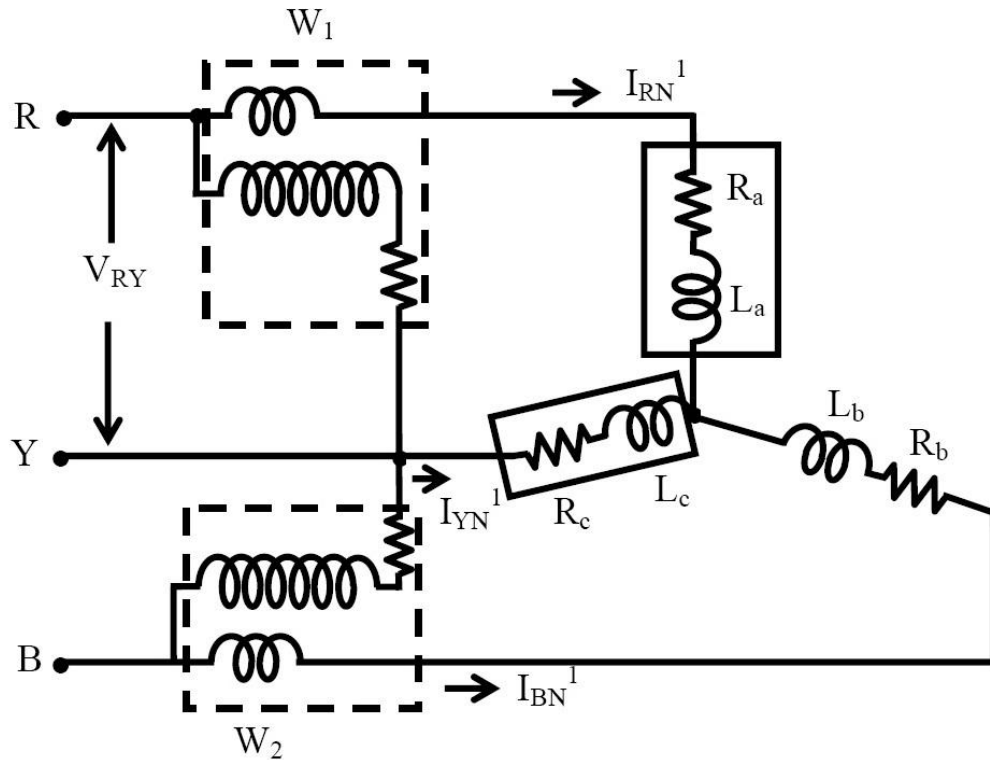


Fig 1: Connection diagram for three phase power measurement using two wattmeter method

The connection diagram for the measurement of power in three phase power measurement circuit using two wattmeter's method is shown in figure 1. This is irrespective of the circuit connection star or delta. The circuit may be taken as balanced or unbalanced one, balanced type being only a special case. Please note the connection of two wattmeters. The current coil of the wattmeter's 1 and 2 in series with R and B phase with the pressure voltage coils being connected across R-Y and B-Y respectively. Y is the third phase in which no current coil is connected.

If star connected circuit is taken as an example the total instantaneous power consumed in the circuit is,

$$W = I_{RN} * V_{RN} + I_{YN} * V_{YN} + I_{BN} * V_{BN} \dots (1)$$

Each of the terms in the above expression equation (1) is the instantaneous power consumed by the phases. From the connection diagram, the circuit in and the voltages across the respective (current, pressure or voltage) coils in the wattmeter, W_1 are I_{RN} and

$$V_{RY} = V_{RN} - V_{YN}$$

So, the instantaneous power measured by the wattmeter W_1 is

$$W_1 = I_{RN} * V_{RY}$$

Similarly the instantaneous power measured by the wattmeter W_2 is

$$W_2 = I_{BN} * V_{BY} = I_{BN} * (V_{BN} - V_{YN})$$

Some of the two readings as given above is,

$$\begin{aligned} W_1 + W_2 &= I_{RN}(V_{RN} - V_{YN}) + I_{BN}(V_{BN} - V_{YN}) \\ &= I_{RN}V_{RN} + I_{BN}V_{BN} - V_{YN}(I_{RN} + I_{BN}) \dots (2) \end{aligned}$$

$$\text{and } I_{RN} + I_{BN} + I_{YN} = 0$$

applying in equation (2),

$$W_1 + W_2 = I_{RN}V_{RN} + I_{BN}V_{BN} + V_{YN}I_{YN} \dots (3)$$

Equation (1) is compared with equation (3) to give the total instantaneous power consumed in the circuit. They are found to be same. The phasor diagram of three phase balanced star connected circuit is shown in figure 2.

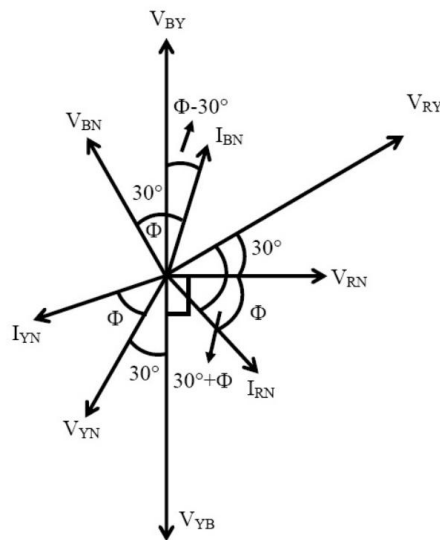


Fig 2: Phasor diagram of three phase balanced star connected circuit

Procedure

BALANCED LOAD:

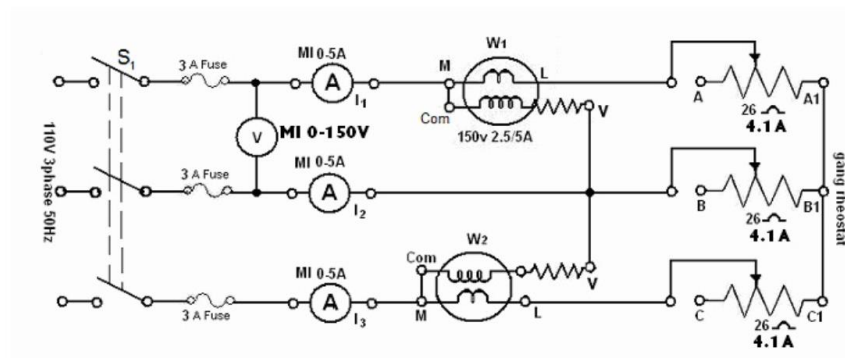


Fig. 1. Three phase power measurement circuit under balance condition

1. Connect the circuit as shown in Fig. 1.
2. Adjust the ganged rheostat for the maximum resistance.
3. Switch on the supply.
4. Close switch S_1 .
5. Read the meters to obtain V_L , I_1 , I_2 and I_3 . Note the wattmeter reading W_1 and W_2 (Note the multiplying factor on the wattmeter).
6. Vary the load resistance and obtain at least five sets of observations, the current should not exceed the limit (4.1 A).

UNBALANCED LOAD:

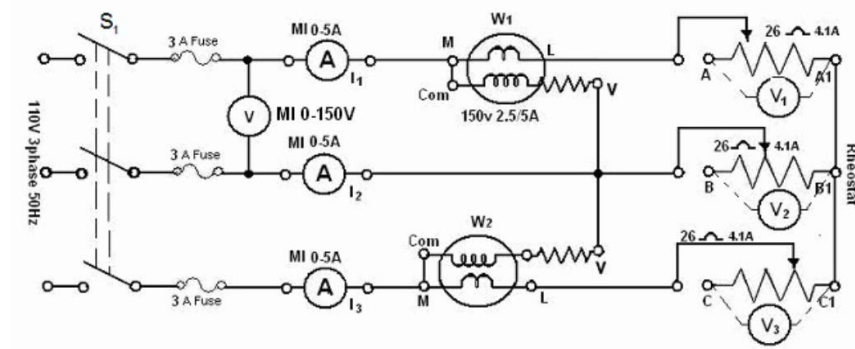


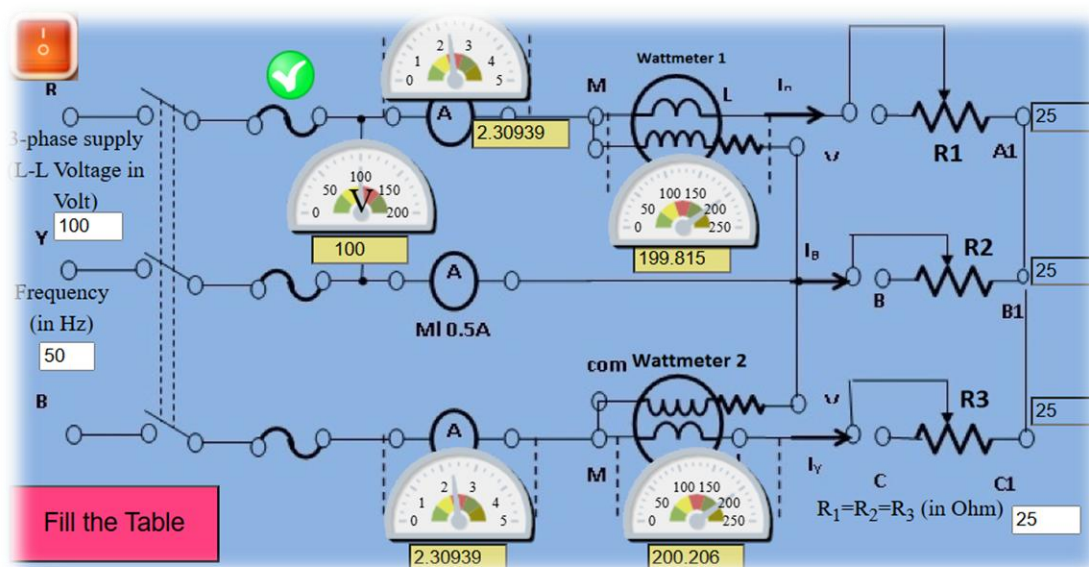
Fig. 2. Three phase power measurement circuit under unbalance condition

- 1) Connect the circuit as shown in Fig. 2.
- 2) Replace the ganged rheostat by three separate rheostats of $26\ \Omega$, 4.1 A and connect in a star.
- 3) Adjust the three rheostats at the maximum values.
- 4) Switch on the supply and set the autotransformer to 110 V.
- 5) Close switch S_1 and take five sets of observation for different rheostat settings such that the reading of I_1 , I_2 and I_3 in each set is appreciably different to create unbalanced loading condition. The current should not exceed the limits in each arm.

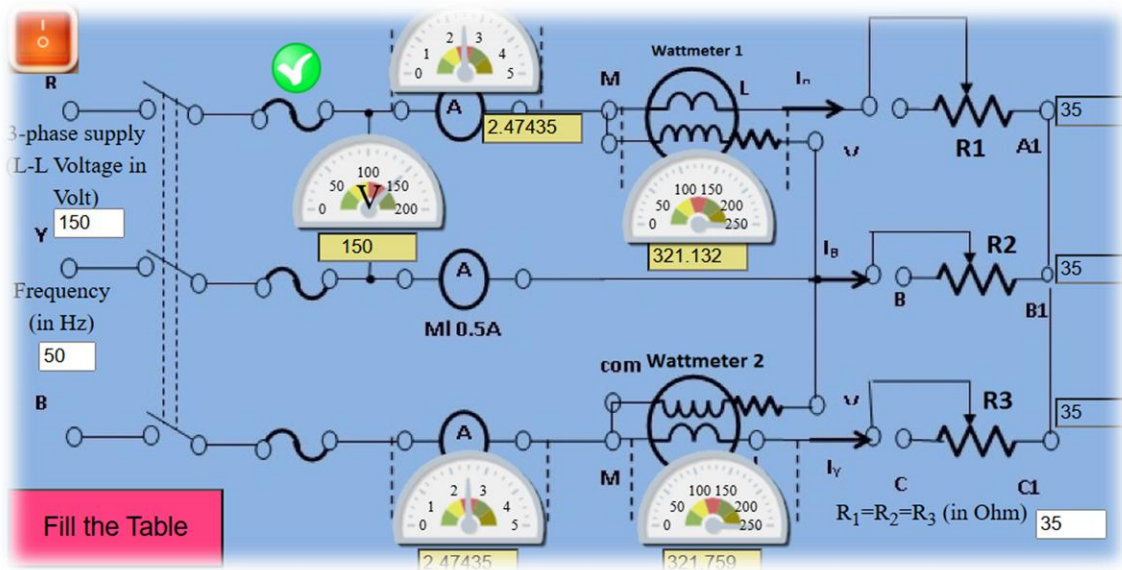
SIMULATION

Balanced

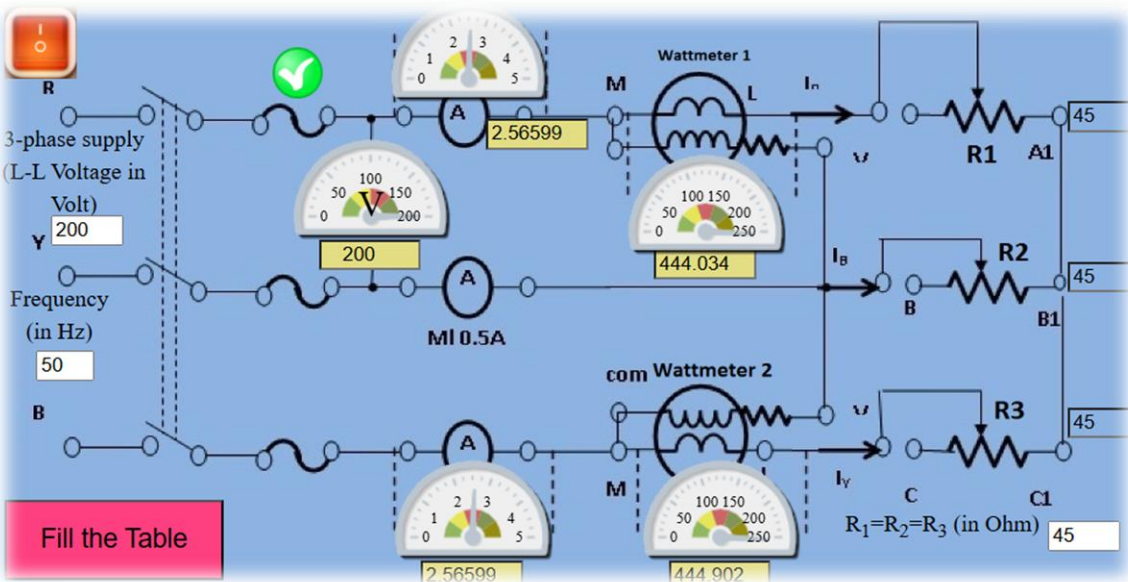
Case 1:



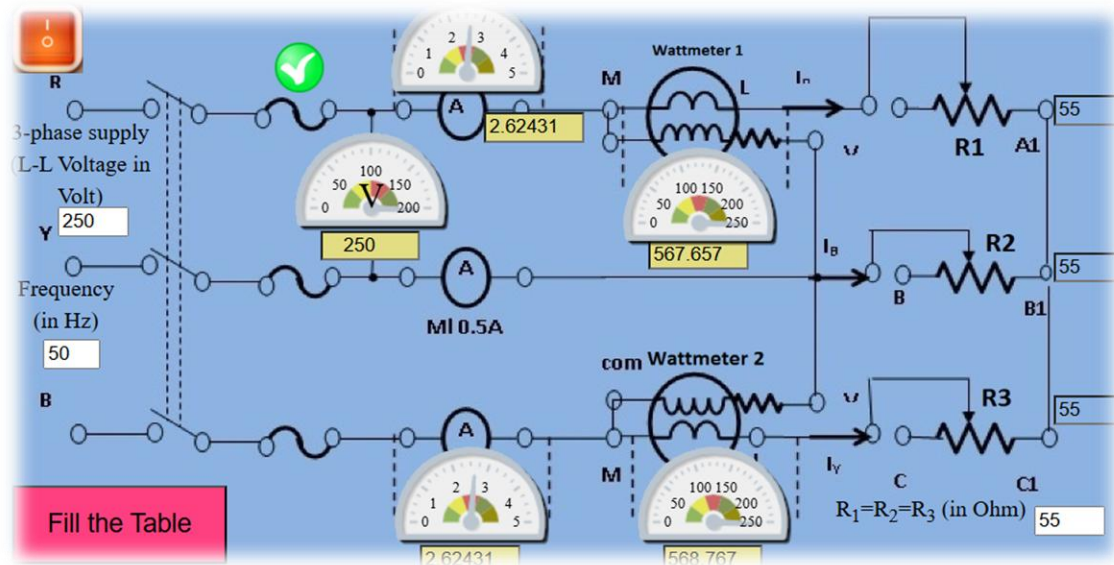
Case 2:



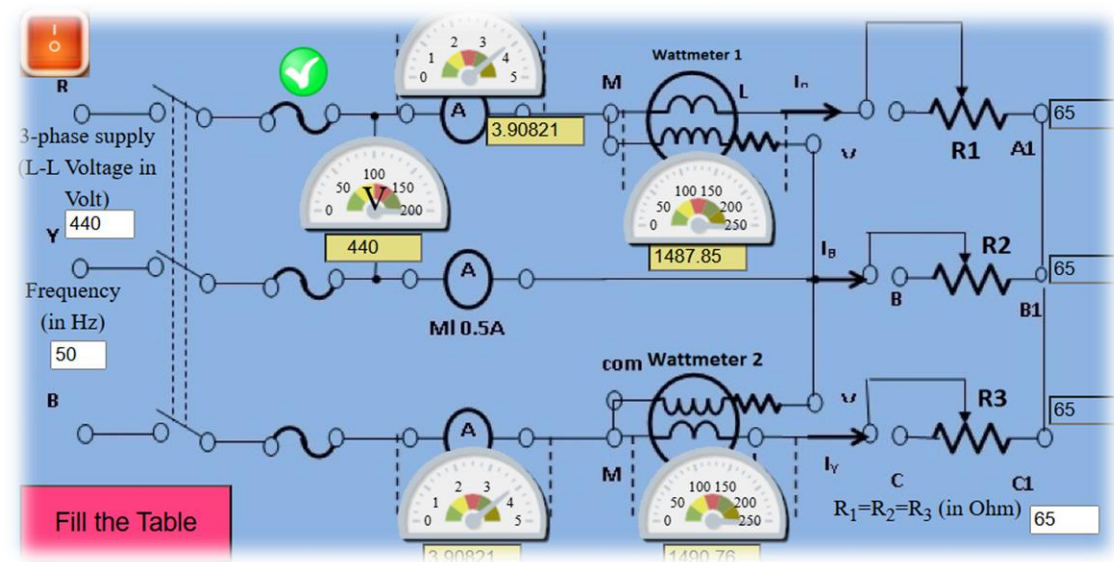
Case 3:



Case 4:



Case 5:

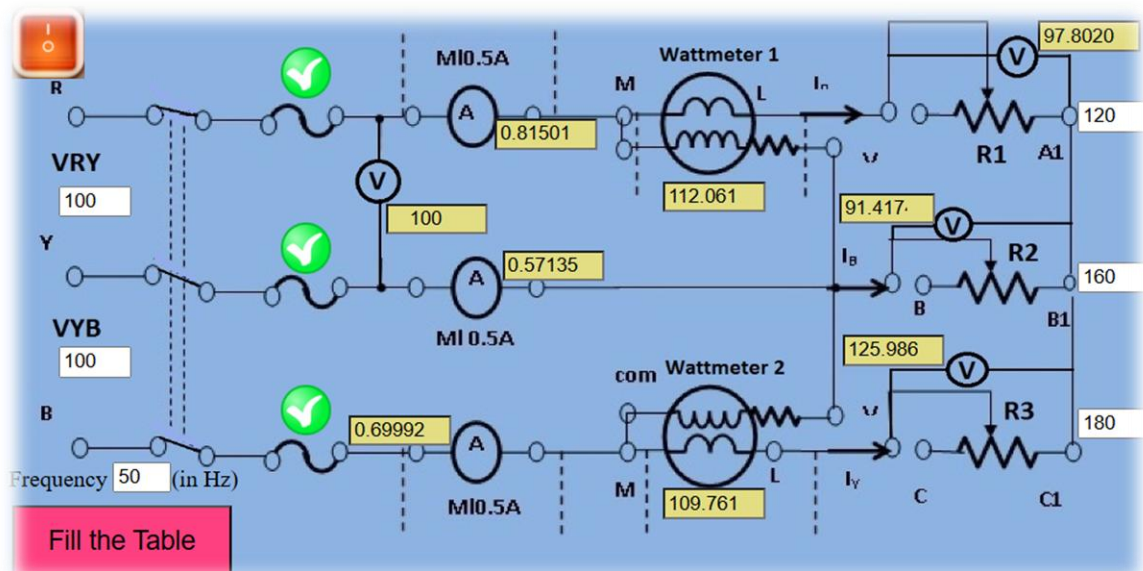


Tabulation for Balanced load:

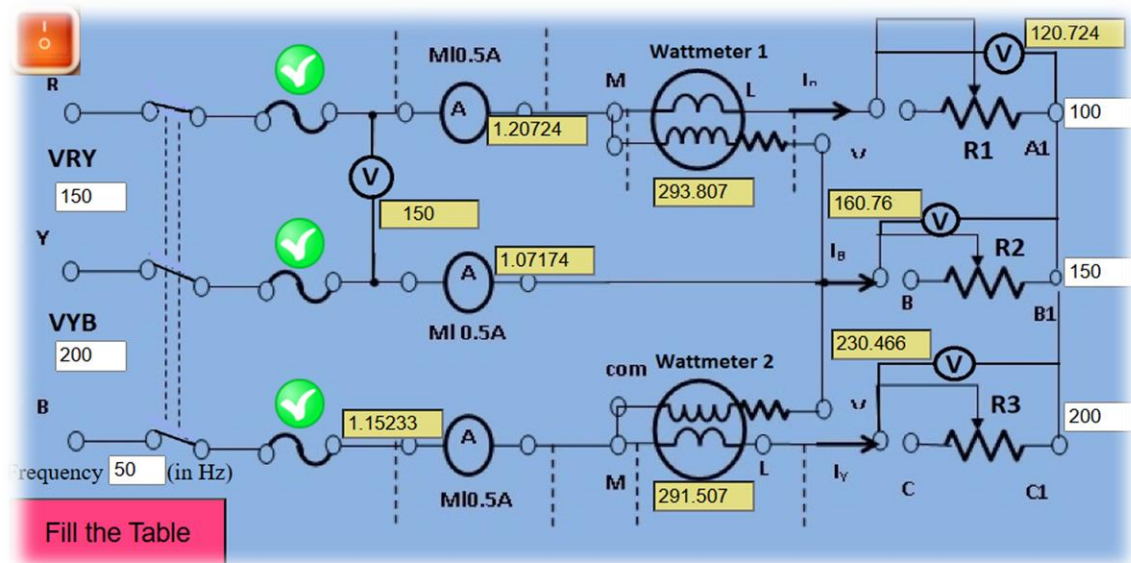
Observation Table											
Serial no. of Observation	V _{RY}	I _R (Amp)	Cos(V _{RY} , I _R)	V _{BY}	I _B (Amp)	Cos(V _{BY} , I _B)	I ₃ (Amp)	W ₁	W ₂	W _C (Calculated power)	W _M (Measured Power=W ₁ +W ₂)
1st	100	2.3093977	0.8652280	100	2.3093977	0.8669190	2.3093977	199.81557	200.20609	399.99885	400.02166
2nd	150	2.4743547	0.8652280	150	2.4743547	0.8669190	2.4743547	321.13216	321.75979	642.85530	642.89196
3rd	200	2.5659975	0.8652280	200	2.5659975	0.8669190	2.5659975	444.03460	444.90243	888.88634	888.93703
4th	250	2.6243156	0.8652280	250	2.6243156	0.8669190	2.6243156	567.65787	568.76731	1136.3603	1136.4251
5th	440	3.9082116	0.8652280	440	3.9082116	0.8669190	3.9082116	1487.8574	1490.7653	2978.4530	2978.6228
6th	440	3.9082116	0.8652280	440	3.9082116	0.8669190	3.9082116	1487.8574	1490.7653	2978.4530	2978.6228
7th	520	5.0542126	0.8652280	520	5.0542126	0.8669190	5.0542126	201.02181	200.10131	4130.3807	4130.4521

Unbalanced

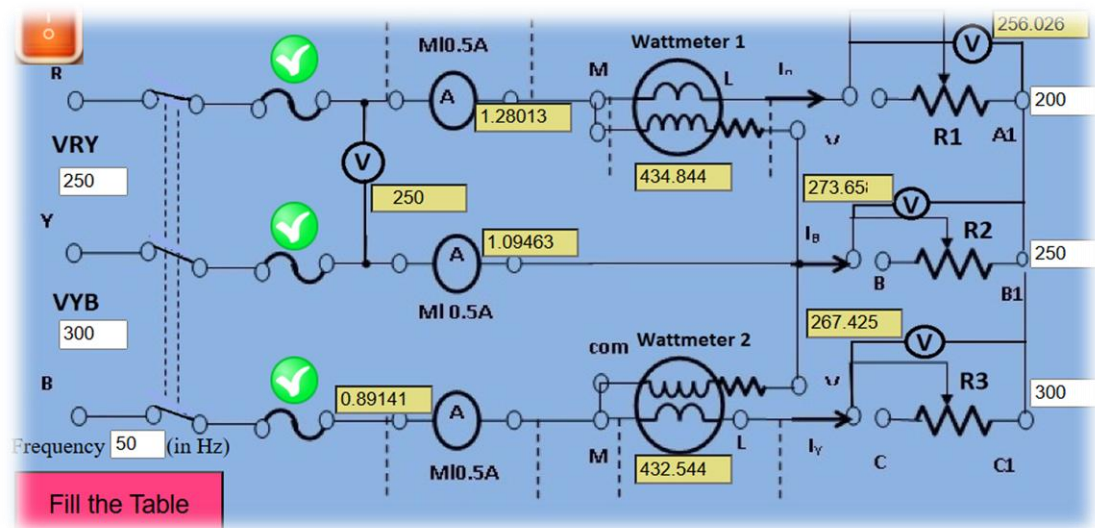
Case 1:



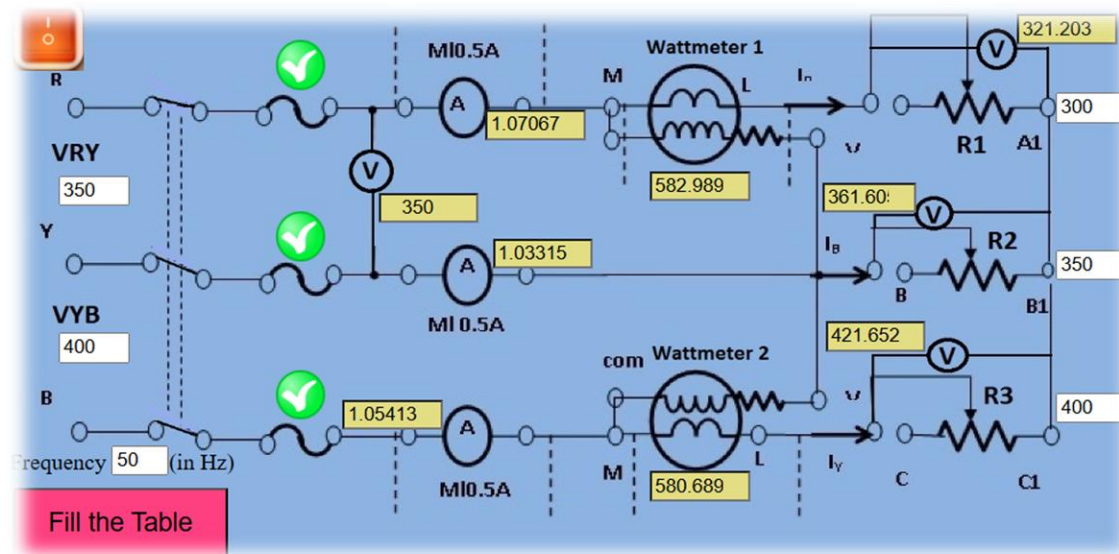
Case 2:



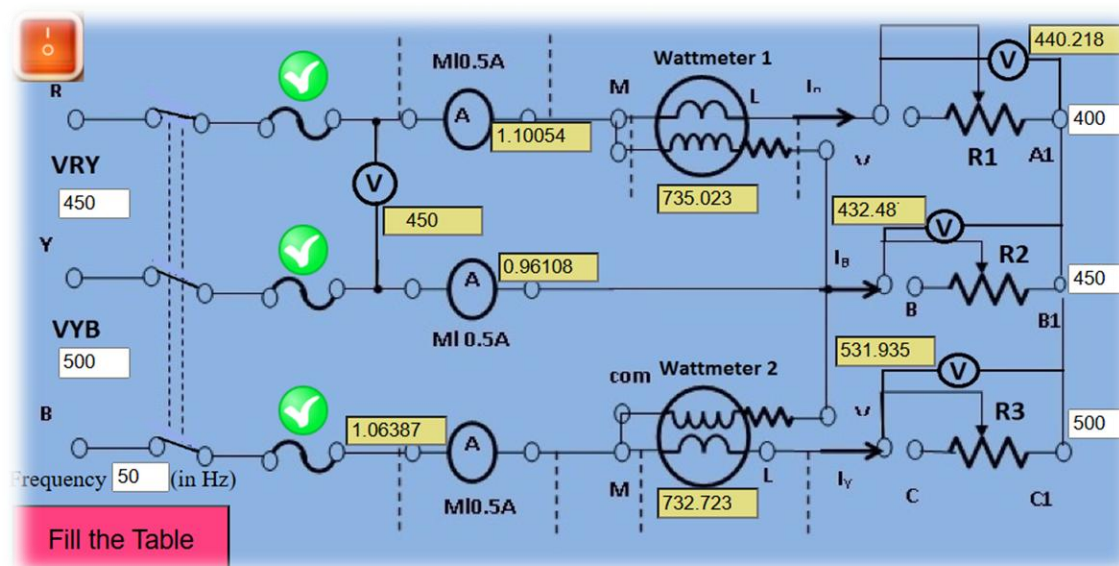
Case 3:



Case 4:



Case 5:



Tabulation for Unbalanced load:

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Observation Table										
Serial no. of Observation	V_R	V_Y	V_b	I_R (Amp)	I_Y (Amp)	I_B (Amp)	W_C (Calculated power)	W_1	W_2	W_M (Measured Power= W_1+W_2)
1st	97.802081	91.417450	125.98662	0.8150173	0.5713590	0.6999256	220.12385	112.06192	109.76192	221.82385
2nd	120.72484	160.76164	230.46644	1.2072484	1.0717442	1.1523322	583.61415	293.80707	291.50707	585.31415
3rd	256.02620	273.65806	267.42532	1.2801310	1.0946322	0.8914177	865.68975	434.84487	432.54487	867.38975
4th	321.20390	361.60530	421.65272	1.0706796	1.0331580	1.0541318	1161.9794	582.98974	580.68974	1163.6794
5th	440.21834	432.48732	531.93519	1.1005458	0.9610825	1.0638702	1466.0467	735.02335	732.72335	1467.7467

RESULT:

Thus the Measurement of Power is Simulated and Validated.

Exp:2.2: Measurement of Energy

AIM:

- i. Understanding the use of Energy-meter.
- ii. In depth knowledge on Energy metering.
- iii. Understanding the calculation of Energy meter error in three types of loading conditions.

THEORY:

Introduction

The Energy Meter is a continuously operating measuring device that displays and records the electric energy consumed over a period by multiplying the measured instantaneous voltage and current. Induction type of energy meters is universally used for the measurement of energy in domestic and industrial a.c. circuits. Induction type of meters possess lower friction and higher torque/weight ratio. Also they are inexpensive and accurate, and retain their accuracy over a wide range of loads and temperature conditions.

There are four main parts of the operating mechanism:

- (i) Driving system
- (ii) Moving system
- (iii) Braking system
- (iv) Registering system

Driving System:

The driving system of the meter consists of two electromagnets. The core of these electromagnets is made up of silicon steel laminations. The coil of one of the electromagnets is excited by the load current. This coil is called the 'current coil'. The coil of second electromagnet is connected across the supply and, therefore, carries a current proportional to the supply voltage. This coil is called the 'pressure coil'. Consequently the two electromagnets are known as series and shunt magnets respectively. Copper shading bands are provided on the central limb. The position of these bands is adjustable. The function of these bands is to bring the flux produced by the shunt magnet exactly in quadrature with the applied voltage.

Moving System:

This consists of an aluminium disc mounted on a light alloy shaft. This disc is positioned in the air gap between series and shunt magnets.

Braking System:

A permanent magnet positioned near the edge of the aluminium disc forms the braking system. The aluminium disc moves in the field of this magnet and thus provides a braking

torque. The position of the permanent magnet is adjustable, and therefore, braking torque can be adjusted by shifting the permanent magnet to different radial positions.

Registering (Counting) Mechanism:

The function of a registering or counting mechanism is to record continuously a number which is proportional to the revolutions made by the moving system.

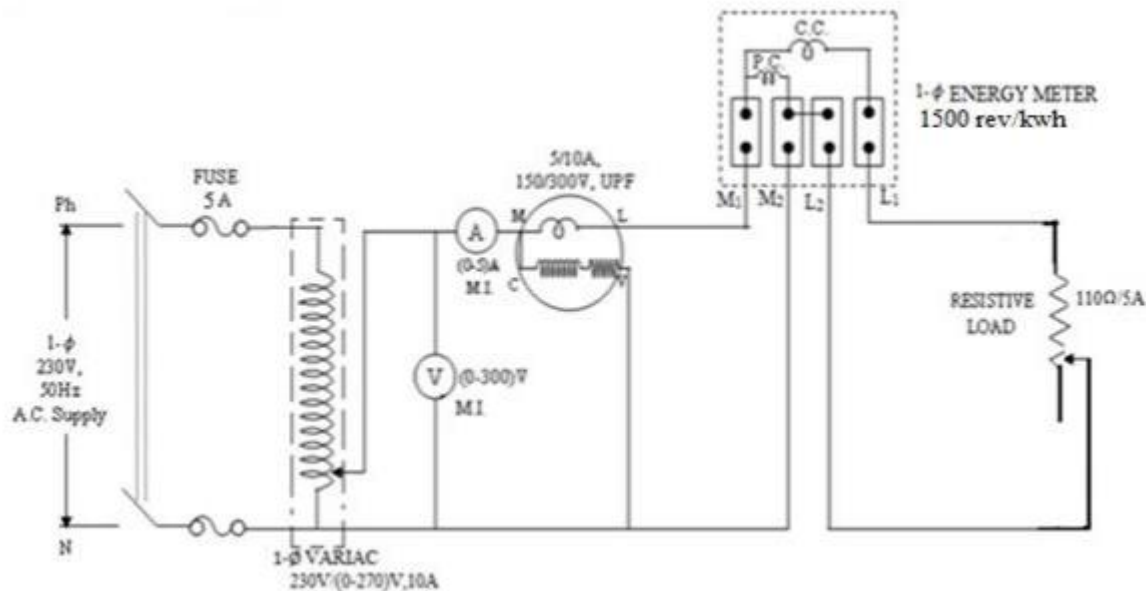


Fig. 1 Calibration and Testing of Single Phase Energy Meter

In all induction instruments we have two fluxes produced by currents flowing in the windings of the instrument. These fluxes are alternating in nature and so they produce emfs in a metallic disc or a drum provided for the purpose. These emfs in turn circulate eddy currents in the metallic disc or the drum. The braking torque is produced by the interaction of eddy current and the field of permanent magnet. This torque is directly proportional to the product of flux of the magnet, the magnitude of eddy current and effective radius 'R' from axis of disc. The moving system attains a steady speed when the driving torque equals the braking torque.

The term testing includes the checking of the actual registration of the meter as well as the adjustments done to bring the errors of the meter within prescribed limits. AC energy meters should be tested for the following conditions:

1. At 5% of marked current with unity pf.
2. At 100% or 125% of marked current.
3. At one intermediate load with unity pf.
4. At marked current and 0.5 lagging pf.

Advantages of Energy Meter:

1. It calculates the electricity consumption quickly.
2. It provides real-time data.
3. It minimises blackout and electric failures.
4. It reduces distribution losses.
5. The electricity meter is energy efficient.

Disadvantages of Energy Meter:

1. The electricity meter can abruptly stop showing readings.
2. If there are power fluctuations, it might show error.

PROCEDURE:

Step 1: Make Connections as per the instructions given below:

1. **S1** of MCB - **A1** of Ammeter and **A2** of Ammeter - **V1** of Voltmeter
2. **V1** of Voltmeter - **Mw** of Wattmeter and **Mw** of Wattmeter - **C** of Wattmeter
3. **S2** of MCB - **V** of Wattmeter and **V** of Wattmeter - **N1** of Energy Meter
4. **N2** of Energy Meter - **L2** of Load and **Le** of Energy Meter - **L1** of Load
5. **L** of Wattmeter - **M** of Energy Meter and **S2** of MCB - **V2** of Voltmeter

NOTE: If wire is wrongly connected, click on node number to detach the wire.

Step 2: Click on **CHECK** button for checking the connections.



Step 3: Click on the  MCB to Turn ON  the supply.

Step 4: Select the number of Bulbs from the Lamp Load.

Step 5: Count the number of times Green LED blinks in 1 min (Meter Counter) by using 'Stopwatch'.

Step 6: Enter the number of times Green LED blinks in 1 min (Meter Counter).

Step 7: Click on **ADD** button to add the readings to the Observation Table.

Step 8: Add different readings to the table by selecting different number of bulbs.

Step 9: Click on **PRINT** button to print the webpage.

Step 10: Click on **RESET** button to refresh the webpage.

FORMULA

Actual Energy (A) = VIT1000 kw-hr

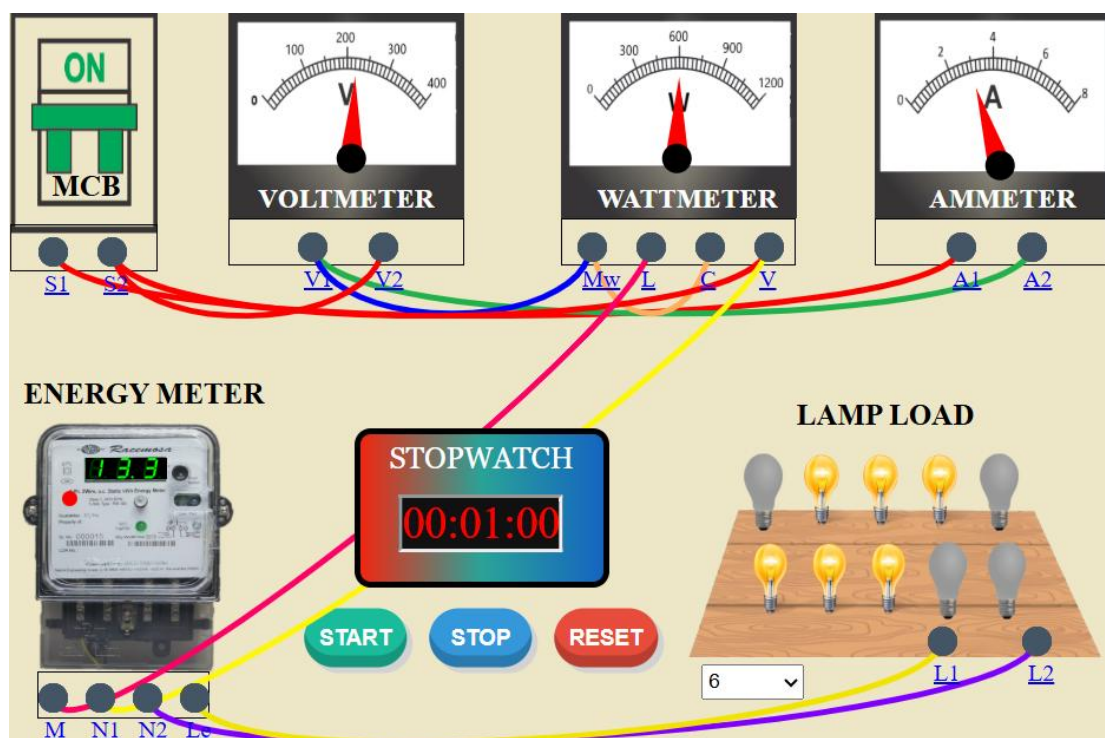
Recorded Energy (R) = Number of Impulse (1min)/Meter Constant

Meter Constant = 3200 Imp/kw-hr

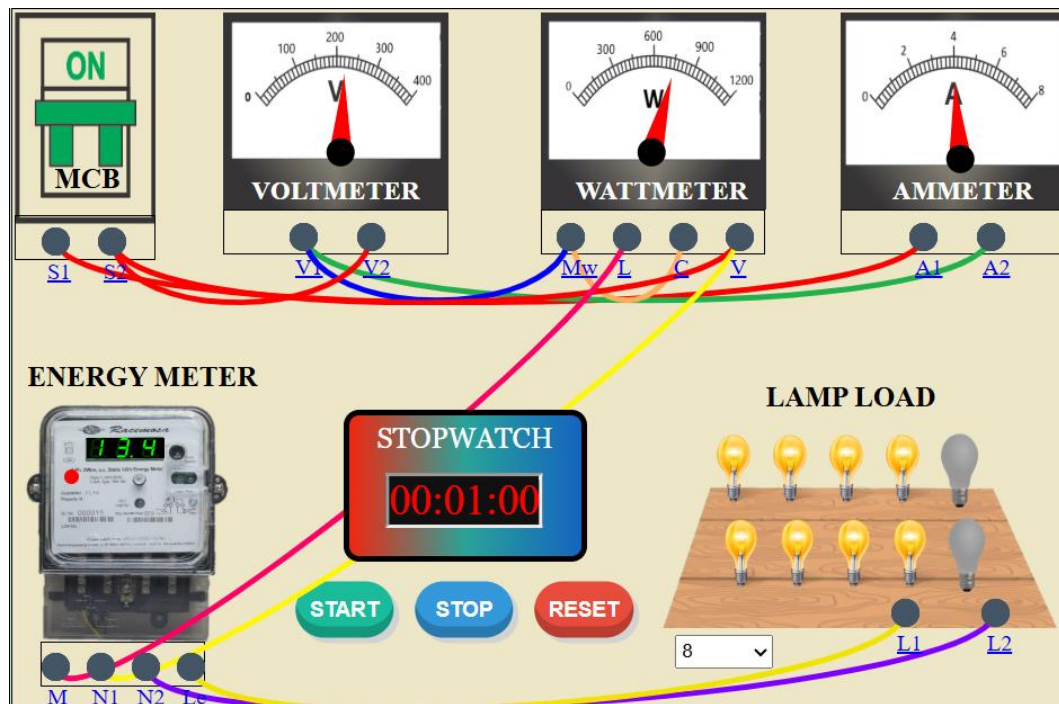
Relative Error = $R - AA * 100\%$

SIMULATION

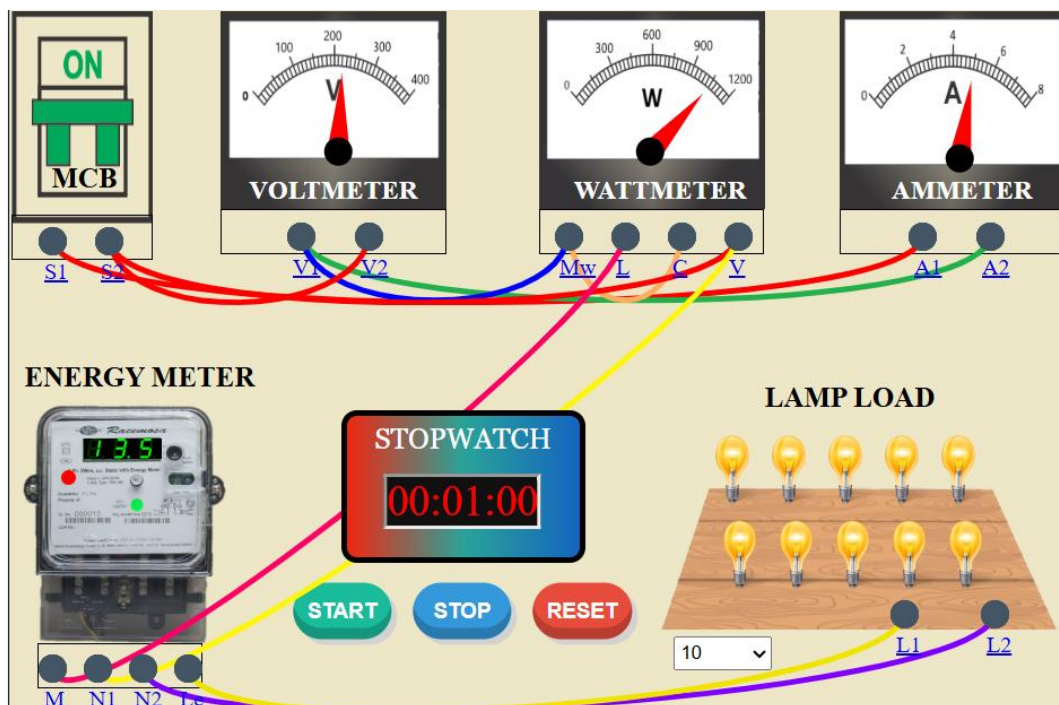
Case 1:



Case 2:



Case 3:



TABULATION:

OBSERVATION TABLE							
S.NO.	LOAD (KW)	VOLTAGE (V)	CURRENT (A)	NUMBER OF IMPULSE	ACTUAL ENERGY (KWh)	RECORDED ENERGY (KWh)	RELATIVE ERROR (%)
1	0.6	220	3	36	0.0109	0.0112	2.75
2	0.8	220	4	48	0.0146	0.015	2.73
3	1	220	4.5	54	0.0165	0.0168	1.81

RESULT:

Thus the Measurement of Energy is Simulated and Validated.